Hands-On Lab (HOL-03): Create a Web Server on AWS (Linux + Windows)

Author : **coderameen**

Learning Objective

After completing this lab, participants will be able to:

- Understand what a web server is
- Configure a Linux EC2 instance as a web server using Apache
- Configure a Windows EC2 instance as a web server using IIS
- Host a simple **Hello / Welcome** web page on both Linux and Windows

---

Learning Outcome

By the end of this lab, you will:

- Clearly understand how web servers work
- Install and manage Apache on Linux
- Install and manage IIS on Windows
- Access hosted web pages using a browser
- Gain confidence in creating real-world web servers in AWS

Introduction: What is a Web Server?

A **web server** is a machine that: - Receives requests from users (browser) - Sends back web pages (HTML content)

In simple words: - Browser asks → "Give me a page" - Web server replies → "Here is your page"

Examples of web servers: - Apache (Linux) - Nginx (Linux) - IIS (Windows)

Session 1: Linux Web Server using Apache

Session 1 Objective

In this session, we will: - Use an existing Linux EC2 instance - Install Apache Web Server - Host a simple **Hello DevOps** page

---

Step 1: Prerequisite

Make sure: - Linux EC2 instance is **running** - Port **22 (SSH)** is allowed - Port **80 (HTTP)** is allowed in Security Group

Why Port 80?

- Port 80 is used for HTTP traffic
- Browsers use this port to access web pages

Step 2: Connect to Linux EC2 Instance

Connect using SSH:

```
ssh -i my-ec2-key.pem ec2-user@<Public-IP>
```

Step 3: Update the System

Before installing anything, update packages:

```
sudo yum update -y
```

Why this step?

- Ensures latest packages
- Avoids dependency issues

Step 4: Install Apache Web Server

Install Apache using:

```
sudo yum install httpd -y
```

What is Apache?

- Apache is a popular web server software
- It listens on port 80
- Serves web pages to users

Step 5: Start and Enable Apache

Start Apache:

```
sudo systemctl start httpd
```

Enable Apache at boot:

```
sudo systemctl enable httpd
```

Why enable?

- Apache will start automatically after reboot

Step 6: Create a Hello Web Page

Apache default web directory:

```
/var/www/html
```

Create index page:

```
sudo vi /var/www/html/index.html
```

Add content:

```
<html>
  <body>
    <h1>Hello, Welcome to the DevOps World!</h1>
```

```
  </body>
</html>
```

Save and exit.

Step 7: Access Linux Web Server

1. Open browser
2. Enter:

```
http://<Public-IP>
```

You should see:

Hello, Welcome to the DevOps World! 🎉

Session 2: Windows Web Server using IIS

Session 2 Objective

In this session, we will: - Use a Windows EC2 instance - Install IIS Web Server - Host a simple welcome page

---

Step 1: Prerequisite

Make sure: - Windows EC2 is running - Port **3389 (RDP)** allowed - Port **80 (HTTP)** allowed in Security Group

---

Step 2: Connect to Windows EC2

1. Use RDP to connect
2. Login as:
3. Username: Administrator

Step 3: Install IIS Web Server

1. Open **Server Manager**
2. Click **Add Roles and Features**

3. Choose **Role-based installation**
4. Select local server
5. Select **Web Server (IIS)**
6. Click **Next → Install**

What is IIS?

- IIS = Internet Information Services
- Microsoft web server
- Used for Windows applications

Step 4: Verify IIS Installation

1. Open browser inside Windows EC2
2. Go to:

```
http://localhost
```

Default IIS page should appear

Step 5: Create Hello Web Page

Default IIS path:

```
C:\inetpub\wwwroot
```

1. Open `index.ht`
2. Replace content with:

```
<html>
  <body>
    <h1>Hello, Welcome to the DevOps World from Windows Server!</h1>
  </body>
```

Save the file

Step 6: Access Windows Web Server from Browser

From your local machine:

```
http://<Windows-Public-IP>
```

You should see the welcome message 🎉

Clean Up (Mandatory)

To avoid billing:

1. Terminate Linux EC2 instance
2. Terminate Windows EC2 instance

Lab Summary

In this lab, you learned:

- What a web server is
- How to install Apache on Linux
- How to install IIS on Windows
- How to host a simple Hello application

End of Hands-On Lab (HOL-03)